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Regional Corruption, Foreign Trade, and Environmental Pollution

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Abstract: As an effective means and an important guarantee for environmental pollution management in China, enhancing the level of foreign openness and reducing the level of regional corruption, respectively, the successful implementation of both are key steps in determining the future development of China's transformation of trade development and green transformation. This paper attempts to systematically examine the mechanisms of regional corruption and foreign trade on environmental pollution from both theoretical and empirical levels. Using the panel data of 30 provinces in China from 2004 to 2017, this study constructs a dynamic panel model with a one-stage pollution index. The system GMM is used to verify the relationship between corruption, trade, and the environment. Empirical results show that corruption reduces investment in environmental governance, R&D, and the introduction of environmental technology, and it increases environmental pollution by reducing the implementation and control of environmental policies. After excluding the effect of corruption on trade, foreign trade is conducive to the improvement of environmental pollution. Meanwhile, the intensification of corruption on pollution will be weakened with increased openness; that is, China's expansion of foreign trade is beneficial to attenuating the pollution effect of corruption on the environment. These findings suggest that expanding trade will ultimately improve the environment and even mitigate the negative impact of corruption on the environment. Therefore, China should dredge the channel of factor flow, give full play to the vitality of market players, strictly investigate corruption, and encourage opening up.

Keywords: corruption; foreign trade; environmental pollution; China; dynamic panel model; empirical testing

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1. Introduction

Since its reform and opening up, China's economy has developed rapidly, and people's living standards have continuously improved. However, China's environment has also been critically damaged [1]. Over the past 20 years, China's GDP has maintained an average growth rate of 10% per year [2], and emissions of some pollutants tripled from 1990 to 2005 [3]. As one of the troikas driving the economy, pollution caused by foreign trade has also affected the country's sustainable development. In 2020, China's total trade volume amounted to CNY 32.16 trillion, accounting for 31.65% of the GDP, and the annual number of deaths from air pollution caused by trade reached 238,000 [4]. Moreover, due to stricter environmental regulations in developed countries, many nations transfer their pollution-intensive industries to China, which also brings more pollutant emissions from commodity production. Pollution-intensive industries still account for a large proportion of China's trade structure and have an increasing yearly trend, which intensifies Chinese environmental pressure [5].

The 18th National Congress of the Communist Party of China proposed that the construction of an ecological civilization should be placed in a prominent position in the overall economic development, paying particular attention to the environmental pollution

that threatens the health of citizens. Accordingly, the Congress proposed a series of environmental policies and industrial policies related to the green economy, such as ecological environmental protection, environmental pollution control, and protective development and economical use of resources. Among them, environmental regulation is considered as an important means to achieve sustainable development and solve the problems of resource shortage and environmental pollution [6]. Multiple policy instruments possess different governance effects and costs [7]. In addition, due to the differences in institutional factors such as the tax sharing system, performance assessment methods, natural conditions, industrial layout, and development status [8], the effects of environmental regulation tool implementation also vary significantly among different regions. The "14th Five-Year Plan" outline also emphasizes the need to continuously improve environmental quality, strengthen pollution prevention and control, and plan economic and social development to ensure the harmonious coexistence of human beings and nature. In terms of trade, it proposes to uphold the green concept and deepen exchanges and cooperation in green development. The public has also increased their awareness of environmental concerns. After 2016, the peaks of the "PM2.5" and "smog" Baidu Index continued to rise. As motor vehicle emissions and traffic congestion have been worsening air pollution [9], daily, an increasing number of people consider the impact of air pollution and opt for public transportation, such as subways and buses, rather than private transportation.

However, the downside of such a scenario is that many companies tend to bribe officials to reduce the cost of environmental regulations. Moreover, we found that the higher the level of corruption, the higher the pollutant emissions [10–12]. Regions such as Jilin Province tend to be relatively less open. Koh [13] pointed out that the increase in openness created more opportunities for corruption. Nowadays, when China pays attention to environmental protection, it should pay more attention to the impact of corruption on environmental pollution. However, this has created a contradiction. The country wants to obtain the economic benefits of openness and to reduce the incidence of corruption. Since the 18th National Congress of the Communist Party of China, the country has introduced several regulations and systems to strictly govern the party, putting the construction of party style and clean government and the fight against corruption in a more prominent position. However, the Global Corruption Perceptions Index released by Transparency International shows that China is a country with more serious corruption. This information has brought the problem of environmental pollution control to the forefront of sustainable economic and environmental development. This scenario is a real dilemma that this study helps address.

This study collects panel data from 30 provinces in China from 2004 to 2017 to explore the relationship between trade and environmental pollution and the moderating effect of foreign corruption on this relationship. The second section of this paper presents a brief summary of the literature, the third section contains the theoretical analysis, the fourth section explains the model setting and variable processing, and the fifth section discusses the policy recommendations.

2. Literature Review

Over the years, the government has promulgated and amended laws and regulations to address environmental pollution in China. Source control is one of the best practices for pollution control. Therefore, stringent law enforcement, public support, and monitoring should be implemented [14]. In addition to the promulgation of policy, policy implementation should be considered. The impartiality of law enforcement is an important guarantee for the effective implementation of policies. As a phenomenon that is widespread across countries worldwide, corruption clearly affects the fairness of law enforcement. It not only causes great harm to the economy and society but also adversely affects the environment in many ways [15–21]. First, officials cannot guarantee the fairness and strictness of law enforcement when they also use public rights to seek private interests. This situation results in the reduction of the effect of environmental regulations [22], increases the risks and

costs of foreign direct investment (FDI) [23], and hinders the spillover effects of green technology on the environment improvement. Second, corruption affects the efficiency of environmental governance by distorting the structure of public expenditures. Corrupt officials will reduce investment in environmental governance, but they invest in areas where economic performance is more significant or bribery is easy to obtain [24]. Moreover, companies seek asylums from the government. Even if they spend a large amount of money and time and lose parts of the resources used for production, they are willing to avoid many environmental regulations or green technology transformation costs and underestimate the social responsibility of protecting the environment [25]. In this process, not only the output of enterprises is affected, but the laws, regulations, and institutional foundations implemented to promote economic development are also shaken due to corruption. This is because corruption has a negative impact on the economic growth and per capita income and even raises the income inflection point of the EKC curve; that is, with the increase in per capita income, environmental pollution has also increased. After reaching a certain "turning point," as per capita income further increases, the degree of environmental pollution gradually slows down [26]. In particular, the increase in income level will stimulate people's demand for environmental protection, thereby encouraging the government to increase investment in environmental governance and improve supervision standards [27]. This implies that corruption promotes environmental pollution by weakening the optimization effect of income growth on the environment. Many studies have found that the relationship between corruption and environmental pollution is largely indirect. For example, many scholars emphasize that corruption has an effect on environmental pollution through indirect variables, such as formulation of legal rules [19,28] and trade liberalization [29]. In particular, corruption affects FDI, green technology, enterprise output, and per capita income, which will also affect foreign trade. The FDI will create new trade opportunities between countries and increase the scale of trade between them [30]. The increase in exports and per capita income indicates a surge in external supply and demand, which has a positive impact on trade. Therefore, as one of the key factors of globalization [31], the impact of corruption on environmental pollution under an open economy has gradually attracted scholarly attention. Additionally, the influence of corruption on foreign trade and environmental pollution may be a vital factor in environmental consequences. However, research on corrupt foreign trade and environmental pollution under the same overall framework is relatively rare. Therefore, we still need to clarify the following two issues from the existing literature: the relationship (1) between foreign trade and environmental pollution and (2) between foreign trade and corruption.

In recent years, owing to the importance and complexity of foreign trade in environmental pollution issues, the relationship between trade and the environment has been a hot issue in the international academic community. From a theoretical and empirical perspective, the impact of trade on pollution is not uniform [32–34]. This situation differs between developed and developing countries [10]. However, the generally accepted transmission mechanisms are as follows. First, the expansion of a country's foreign trade leads to an increase in pollution emissions in the use of energy, industrial activities, transportation activities, and resource surveys. Furthermore, foreign trade affects people's environmental impact by changing income levels, environmental protection awareness, and importance [35], which has an impact on environmental governance. Second, trade liberalization will change the flow of resources and output composition of a country based on its comparative advantages. The distribution patterns of polluting products and clean products have a greater impact on pollution. For example, the cost of polluting resources in a country is lower, and environmental regulations are relatively loose; thus, expanding exports will increase pollution. Third, foreign trade brings more advanced clean technology, equipment, and corporate organization and management systems, which can directly improve the country's environmental governance capabilities, environmental awareness, and environmental standards and reduce pollution.

Gil-Pareja [36] proposed that corruption has a positive impact on trade when it plays a role in a highly regulated economy with a poor institutional environment as a deregulation mechanism. However, the opposite conclusion is that corruption will hinder trade. In countries or regions with high levels of corruption, commercial bribery will induce government officials to create more administrative controls and rules to increase opportunities for corruption [37]. For example, when customs officials must be bribed to conduct trade smoothly, the time the goods stay in the port increases [38]. It also implies that these regions often have high trade barriers [39]. After high trade barriers reduce the competition between domestic enterprises and foreign countries, they provide more rent-seeking behaviors for domestic enterprises, which has caused corruption to significantly reduce the trust of other countries and the efficiency of trade and raise the risks and costs of trading partners [37]. Compared with that in developed countries, corruption in developing countries hinders trade more clearly [40]. Conversely, Trofimchuk [41] believes that corruption is directly related to a country's investment level and trade attractiveness, and FDI may also aggravate the level of corruption in transitional countries [42]. Corruption in one country may affect corruption in other countries, and international trade may promote this transnational spillover effect [43].

The literature discussed shows that the impact of foreign trade on environmental pollution depends on the degree of corruption. First, the negative impact of high corruption on economic growth and per capita income will widen the domestic income gap, which results in insufficient domestic demand, reduces import demand for foreign products, and negatively affects total trade. Although this condition may reduce the scale of pollution emissions, people with low-income levels have less awareness of environmental protection and weaker government supervision. The decline in the demand for environment-friendly products will be reflected in the reduction of the cost of production factors of polluting products. Hence, the industrial structure and then the export structure, which has an impact on pollution emissions, may be affected. Second, high trade barriers created by high corruption hindering trade directly reduce the number of advanced green equipment and technologies that are exposed to trade, which may impede the improvement of the country's environmental governance capabilities. Moreover, as mentioned previously, corruption distorts the structure of public expenditures and reduces investment in environmental governance, research and development, education, and other fields. This will also reduce the spillover effect of green technology in the process of foreign trade.

It can be seen that some scholars have studied the role of corruption and trade on the environment, while very little direct reference has been made to the impact of corruption on the environmental pollution and its interaction effects of a country's foreign trade. Most studies have shown that environmental regulations were strengthened by the expansion of trade and weakened by the deepening of corruption [29]. Owing to the rising level of corruption, government policies ignore the consideration of public welfare and encourage the bribery of interest groups [15]; then, the positive effects of trade liberalization will be weakened. Chang [44] also confirmed that corruption reduces the effect of trade on environmental conservation. Therefore, this study contributes to the following aspects. Firstly, we analyze the mechanism of corruption's effect on the environmental pollution and its interaction effect of foreign trade, empirically investigating how different levels of corruption affect the environmental pollution effect of regional foreign trade. Secondly, we consider the non-linear and dynamic connection of corruption to the environment. Thirdly, with regard to regional heterogeneity and differences over time, most studies have focused more on heterogeneity between countries, and few have compared regions within a country. We also examine changes in the quality of a country's environment following important environmental policies. This is important and necessary for the analysis of regional corruption and environmental governance.

3. Model Construction and Indicator Selection

Based on the previous analysis, two core influencing factors, the scale of foreign trade and the level of corruption, are selected to estimate their effects on environmental pollution, and the empirical equations are set as follows.

(1) Static model setting

$$\ln EP_{it} = \alpha_{it} + \beta_1 \ln T_{it} + \beta_2 \ln CO_{it} + \sum \beta_j \ln X_{it} + u_{it} \quad (1)$$

(2) Dynamic model setting

To avoid biased estimates due to the omission of other important explanatory variables, we extend Equation (1) to a dynamic model by introducing a lagged term for the level of environmental pollution emissions, thus being able to eliminate the endogeneity problem of the model through a dynamic panel approach. Simultaneously, considering that any economic activity itself has a certain inertia [45], there is likely to be a lagged effect of pollution emissions in each province. Therefore, the dynamic model is considered to be more consistent with the Chinese reality. The lagged one-period dynamic panel model constructed in this paper is as follows.

$$\ln EP_{it} = \alpha_{it} + \beta_0 \ln EP_{i,t-1} + \beta_1 \ln T_{it} + \beta_2 \ln CO_{it} + \sum \beta_j \ln X_{it} + u_{it} \quad (2)$$

Here, I and t represent province and time, respectively; the dependent variable is selected as EP as the level of environmental pollution, which can be divided into industrial wastewater discharge ($\ln EP_1$), industrial waste gas emissions ($\ln EP_2$), and industrial waste solid emission ($\ln EP_3$) in per capita form; $\ln EP_{i,t-1}$ represents the lagged period of the environmental pollution. T is the scale of foreign trade in each province; CO is the level of corruption in each province; and X_{it} represents other control variables, including economic scale, industrial structure, technological progress, environmental regulations, and FDI. Further, β_1, β_2, and β_j are the regression coefficients of the variables, and α_{it} and u_{it} represent the intercept and random disturbance terms of the model, respectively.

This study selected China's inter-provincial data for research. The data were obtained from the China Economic and Social Big Data Research Platform, China National Bureau of Statistics, China Environmental Yearbook, China Statistical Yearbook, statistical yearbooks of various regions, and China Prosecution Yearbook, with reference to many factors (e.g., the characteristics of the research objects in this paper) and the availability of data integrity, among others. In order to obtain more convincing and covering results, data from 30 provinces (municipalities and autonomous regions) in China from 2004 to 2017 were selected for research and analysis. Tibet, Hong Kong, Macao, and Taiwan were excluded in the analysis sample due to serious data deficiencies. The variables were then logarithmized. Each index of the regression equation is described as follows.

3.1. Environmental Pollution Level (EP)

Commonly used indicators of the level of environmental pollution are the single environmental pollution indicator, such as the total emissions of three industrial wastes, PM2.5, or multiple environmental pollution indicators and composite environmental pollution indicators (i.e., comprehensive consideration and calculation of various emissions). This study selects industrial "three wastes" emissions that cover a wide range of pollutants, causing a greater degree of harm to the human body, and are valued in the national environmental comprehensive management process to reflect the degree of environmental pollution in various provinces in China. Generally, the scale of pollution has continuity in time. If the pollution level in the previous period of inspection is high, the factors that cause pollution will not change significantly in a short time. Therefore, the pollution level in the current period has a positive correlation with the previous period. This aspect makes the predictive coefficient of the lag term of this indicator positive.

3.2. Scale of Foreign Trade (T)

This study uses the total import and export volume of each region to measure the scale of foreign trade. The expansion of a country's foreign trade has led to changes in the scale of production and consumption, increased the country's total resource utilization and pollution emissions, and ultimately led to increased pollution. However, exporting to countries with strict environmental standards encourages enterprises to increase investment in green technology transformation, while imports from high green technology levels have brought advanced emission reduction equipment and production capacity, thereby improving environmental quality. As analyzed above, the impact of foreign trade on the environment is complex, so the predictive coefficient of this indicator cannot be determined.

3.3. Corruption Level (CO)

As this study uses provincial data, and provinces lack the measurement of specialized corruption indicators such as the Corruption Control Index and the Global Corruption Perception Index, the ratio of the total number of corruption cases to the number of public officials is used to construct the corruption index at an object level. From the analysis shown above, corruption not only acts on the dimensions of policy formulation, implementation, and strictness to directly distort the effect of environmental governance but also reduces the efficiency of public investment by affecting economic growth rates, eroding private investment, and indirectly deepening income disparity and social instability. From this perspective, the predictive coefficient of this coefficient is positive.

3.4. Economic Scale (VG)

Economic scale is measured by the per capita GDP of the region. Some studies point out that when social and economic factors effectively curb pollution, economic individuals will grow around the inverted U-shaped Environmental Kuznets Curve [46]. As Grossman and Krueger [47] first proposed in their working papers, with the increase in per capita income, pollution emissions increase with the expansion of economic activities, but with the increase in people's environmental awareness and green technology, environmental regulations have gradually become stricter. Pollution emissions gradually decreased. In other words, environmental quality and economic development present an inverted U-shaped curve. To verify this hypothesis, this article introduces the quadratic of the regional per capita GDP, and the predictive coefficient of this indicator is positive.

3.5. Industrial Structure (IS)

Industrial structure is measured by the output value of the secondary industry in each region as a percentage of the regional GDP. In general, secondary industries, especially industrial manufacturing processes, consume more energy and produce more pollutants than other industries, and the problem of pollution is more prominent. However, China's extensive production methods with high pollution, high consumption, and low output have aggravated this pollution effect. Therefore, the optimization and upgrading of the industrial structure, that is, the decline in the proportion of the primary and secondary industries in the GDP, can reduce the level of environmental pollution, so the predictive coefficient of this indicator is positive.

3.6. Technological Progress (TE)

The capital-to-labor ratio is used to measure technological progress. Technological progress means improvement in resource utilization efficiency and the ability to reduce pollution and emissions, which is conducive to reducing pollution emissions [48] and greatly enhancing the environment. Munir and Ameer [49] also confirm this conclusion. On the one hand, the higher value of the capital–labor ratio means that the products in production have higher capital and technology content, thereby showing a capital-intensive industrial model. Capital-intensive products have a higher technical content than labor-intensive products [50]. On the other hand, they reflect the high technical efficiency of the

industry and the sufficient funding base for technological transformation. The material capital stock is using the perpetual inventory method. For this indicator, the predictive coefficient is negative.

3.7. Environmental Regulation (EI)

Accurate measurement of environmental regulation intensity is still a challenge for current domestic and international research, and the indicators adopted by existing studies vary widely. Among them, the problem of using governance cost to measure EI is that the larger fee means more pollution or more pollution should imply more fees collected. Therefore, this study refers to Chen et al. [45] and uses the frequency of words related to environmental protection in local government work reports as a proxy variable for environmental governance (ER), which is a good indicator of local governments' determination and concern to improve the environment. It is also staggered with the implementation of environmental management throughout the year, which alleviates endogenous problems, with a positive prognostic coefficient.

3.8. Foreign Direct Investment (FDI)

Foreign direct investment (FDI) is measured by the percentage (%) of the actual utilization of foreign capital in each region in GDP. The influx of foreign capital has expanded the scale of domestic production and may also bring about a large number of polluting industries, which will intensify the pressure on my country's resources. Simultaneously, the host country has reduced environmental standards to attract foreign investment and increase pollution emissions. However, the process of introducing foreign capital is accompanied by the agglomeration of industry, and transfer and overflow of advanced clean production technology and management experience, which improves the technological innovation capabilities of enterprises and indirectly reduces industrial pollutant emissions [51]. Therefore, FDI has a greater impact on China's environment, but the specific direction is uncertain, and the predictive result of this indicator is uncertain.

Before analyzing the empirical fitting model of environmental pollution, a descriptive statistical analysis was performed on the full sample data to further understand the characteristics of the research data (Table 1). Table 1 contains the extreme values, the mean, and the standard deviation of the full sample data in logarithmic form, and the last column shows the predicted sign based on the previous theoretical analysis.

Table 1. Descriptive statistical analysis of all variables.

	Variable	Mean	Max	Min	Std. Dev.	Obs	Expected Coefficient
Per capita industrial wastewater discharge	ln*EP1*	2.60	3.86	1.02	0.55	420	/
Per capita industrial waste gas emissions	ln*EP2*	1.25	3.25	−0.44	0.65	420	/
Per capita industrial waste solid emissions	ln*EP3*	0.44	3.40	−2.00	0.85	420	/
Regional GDP per capita	ln*VG*	10.25	11.83	8.35	0.66	420	+
	lnVG^2	105.44	140.00	69.78	13.56	420	−
Industrial structure	ln*IS*	3.76	4.13	2.83	0.21	420	+
Capital labor ratio	ln*TE*	2.71	4.31	0.51	0.75	420	−
Sewage fee income	ln*EI*	0.23	1.46	−1.06	0.45	420	−
Level of corruption	ln*CO*	−1.45	−0.73	−3.08	0.34	420	+
Total import and export	ln*T*	17.15	20.91	12.93	1.67	420	?
Per capita import and export	ln*T*	8.98	12.22	5.93	1.46	420	?
Foreign direct investment	ln*F*	−5.24	−2.72	−9.13	1.12	420	?

Note: ? means the predictive coefficient is uncertain.

4. Empirical Analysis

Considering that the introduction of the first-order lag term of the explained variable and the random error term are likely to have a correlation, and the extensive economic development mode produces a large amount of environmental pollution and resource consumption, effects on the total GDP are short term. In addition, the extensive economic development mode produces a large amount of environmental pollution and resource consumption, which has a significant effect on GDP in the short term. If environmental pollution increases, it may affect regional GDP; thus, there is endogeneity between the two variables. Therefore, to address this endogeneity, the system GMM is used for analysis. Regression results (1) did not include corruption variables and independently tested the impact of foreign trade on environmental pollution. To verify the impact of corruption on environmental pollution, we added corruption variables lnCO on the basis of regression results (1) to obtain regression results (2); to explore how the impact of foreign trade on environmental pollution depends on the degree of corruption, regression results (3) add the corruption variable lnCO and its interaction with foreign trade ln$T$ × lnCO. The results are shown in Table 2.

Table 2. System GMM regression results.

	Per Capita Industrial Wastewater Discharge			Per Capita Industrial Waste Gas Emissions			Per Capita Industrial Waste Solid Emissions		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
$L.\ln EP_{i,t-1}$	0.910 *** (15.83)	0.902 *** (37.80)	0.638 *** (7.25)	0.913 *** (22.97)	0.881 *** (21.24)	0.991 *** (24.64)	0.663 *** (22.54)	0.651 *** (19.98)	0.341 *** (10.52)
lnVG	1.267 (1.16)	0.029 (−1.12)	1.246 (0.69)	0.428 (0.71)	0.579 (0.98)	0.998 * (1.65)	3.079 * (1.92)	4.77 * (1.83)	4.091 (1.62)
lnVG^2	−0.058 (−1.10)	0.031 (2.43)	−0.056 (−0.67)	−0.014 (−0.48)	−0.022 (−0.81)	−0.043 (−1.53)	−0.106 (−1.37)	−0.189 (−1.48)	−0.154 (−1.28)
lnIS	0.120 (−0.47)	0.137 *** (2.43)	0.541 *** (2.54)	0.200 (2.67)	0.213 *** (2.77)	0.193 ** (2.19)	0.654 *** (4.72)	0.655 *** (4.48)	1.059 *** (−1.45)
lnTE	−0.054 (−0.48)	−0.024 (−0.60)	−0.344 (−1.59)	−0.043 (−0.75)	0.001 (0.02)	−0.094 * (−1.73)	−0.378 *** (−3.89)	−0.240 *** (−3.45)	−0.213 (−1.45)
lnEI	−0.026 (−0.77)	−0.022(−0.85)	−0.030 (0.56)	−0.015 (−0.42)	0.009 (−0.34)	−0.008 (−0.26)	0.039 (1.62)	0.026 (0.84)	0.052 ** (2.02)
ln$T$	−0.003 (−0.20)	−0.011 (−1.59)	−0.138 (−1.60)	−0.057 *** (−2.95)	−0.096 *** (−3.03)	−0.149 *** (−2.77)	−0.248 *** (−9.86)	−0.377 *** (−8.42)	−0.442 *** (−4.33)
lnCO		0.070 ** (0.92)	2.265 *** (2.91)		0.002 (0.05)	1.209 ** (2.30)		0.024 (0.80)	1.362 * (1.70)
ln$T$ × lnCO			−0.123 *** (−2.82)			−0.072 ** (−2.37)			−0.079 * (−1.75)
ln$F$	0.035 *** (3.97)	0.037 *** (3.23)	−0.001 (−0.006)	0.023 (1.63)	0.021 (1.44)	0.032 ** (2.29)	0.031 *** (3.88)	0.031 *** (3.98)	0.031 *** (3.32)
cons	−5.940 (−1.17)	2.609 (1.05)	−4.756 (−0.49)	−2.381 (−0.74)	−3.108 (−1.00)	−3.395 * (−1.15)	−17.027 ** (−2.01)	−25.926 * (−1.89)	−20.828 * (−1.55)
AR(2)-p value	0.690	0.676	0.954	0.336	0.347	0.381	0.384	0.381	0.471
Sargan-p value	0.395	0.391	0.141	0.172	0.237	0.286	0.744	0.775	0.647

Note: *, **, and *** represent the significance level of 10%, 5%, and 1%, respectively. The values of the z-statistic are in parentheses. AR(2) is used to judge whether the estimated residuals have a serial correlation. The Sargan test assesses whether the instrumental variables are effective overall.

In Table 2, the regression results of regression results (1) show that, when corruption variables are not introduced, foreign trade improves environmental pollution. The variable corruption is introduced in regression results (2), and the regression coefficient under the indicator of per capita wastewater discharge is significantly positive, indicating that corruption may reduce the implementation and control of environmental policies and reduce investment in environmental governance, R&D, and introduction of environmental technologies, thereby aggravating environmental pollution. The indicators of per capita waste gas and per capita solid waste show that corruption improves environmental pollution, but the results are not significant. In regression results (3), the interactive variables of foreign

trade and corruption were further introduced, and it was found that the coefficient of trade on environmental pollution remains negative. This result shows that after excluding the effect of trade on corruption, corruption makes environmental pollution more severe; further, the regression coefficient of the interaction variable is also significantly negative, indicating that the higher degree of corruption is beneficial to trade in improving the environment.

In the model, the environmental effects of foreign trade have several positive impacts. First, because the expansion of export scale is accompanied by an increase in foreign exchange reserves, it further enhances the import capacity of a large number of advanced technologies, equipment, and superior products and services. This type of foreign exchange reserve transforms the country's actual productivity and realizes capital accumulation. From the income structure perspective, foreign trade has increased China's employment and per capita income, leading to an increase in people's demand for a clean environment. Second, it also led to a rise in China's environmental standards and a decline in environmental pollution. Recently, owing to the upgrading of the commodity structure of China's exports of goods and services, the proportion of exports of primary products in goods exports has declined, the proportion of exports of high-tech products has increased, the proportion of traditional service trade in service exports has declined, and the proportion of emerging services exports has increased significantly. The upgrading of the export commodity structure reduces environmental pollution. Moreover, the expansion of foreign trade and the establishment of foreign green barriers and technical trade barriers have intensified competition among enterprises and stimulated their innovative vitality. The pursuit of innovation in this process reduces the intensity of resource consumption and overall cost of product production, optimizes the domestic supply and demand structure, and enhances the positive side of the structural effect on environmental quality. Together with the improvement of China's R&D level, human capital, and development of finance, the absorption capacity of foreign trade technology spillovers has increased, further improving the environmental quality. The pursuit of innovation in this process reduces the intensity of resource consumption and the overall cost of product production. Hence, the domestic supply and demand structure are not only optimized, but also the positive side of the structural effect on environmental quality is enhanced. Together with China's improved R&D level, increased human capital, and financial development, this has increased the ability to absorb technology spillovers from foreign trade, further improving the quality of the environment.

The empirical test of regression results (3) shows that the coefficient of corruption on environmental pollution is significantly positive, while the regression coefficient of the interactive variables of corruption and foreign trade is significantly negative. This result reveals that higher levels of corruption can better play the role of trade in improving environmental pollution, which is obviously inconsistent with the analysis of the literature review. An increase in the level of corruption has weakened the positive effect of foreign trade on environmental improvement through channels such as the negative impact on per capita income, creation of high trade barriers to hinder trade, distortion of public investment, reduction of environmental governance, and efficiency of green innovation. However, according to Beck and Maher [52] and other scholars, corruption can save time and cost and improve the efficiency of resource allocation; that is, the tedious customs procedures are simplified, and the efficiency of the passage is increased by bribing customs officials. Moreover, the positive and negative effects of corruption are related to the degree of perfection of the regional system. When corruption is used as a deregulation mechanism in a highly regulated economy with a poor institutional environment, it will have a positive impact on trade. When institutional defects exist in the country, corruption will instead play a "grease the wheels," reduce the institutional friction of investment and transactions, and have a beneficial impact on opening up [36]. Therefore, the positive role of corruption in improving the effect of trade on the environment may not be absolute. This result is affected by the comprehensive influence of different factors (e.g., region, time, and system). It can only show that, under the current level of system perfection in China, a certain

degree of corruption may be conducive to the improvement effect of trade on pollution. Nevertheless, with the improvement of the institutional framework (e.g., deepening of democratization and increase in market liquidity), the interaction of corruption and trade on pollution may be negative. Officials may also be motivated to accept more bribes or be affected by the performance appraisal system of local officials to expand foreign trade. The expansion of foreign trade may increase the degree of competition in the domestic market, and lowering environmental standards is no longer necessary to obtain the same output. To the previous level, the "welfare effect" of residents considered by the government to stimulate output and increase consumer surplus has expanded, and it impacts the environment by strengthening the management of transaction permits, environmental taxes, and environmental regulations. Moreover, when corruption becomes a trade barrier (i.e., officials formulate strict rules and regulations on imported goods for private interests, such as discriminatory government procurement policies, and harsh technology and environmental protection standards), it indirectly restricts the import of goods. However, trade barriers have also increased the green technology content of imported products and cultivated the habit of domestic nation's green consumption, which invisibly play a role in demonstrating and supervising domestic products (e.g., popularity of fluorine-free air conditioners, cloth shopping bags, and energy-saving lamps).

The regression coefficients of trade on environmental pollution and those of interactive variables of corruption and foreign trade are all significantly negative. This result shows that the increase in pollution caused by corruption weakens with the increase in the level of openness. This indicates that China's expansion of foreign trade is beneficial to improving the effect of corruption on the environment. Based on the regression coefficients, this study calculates the critical values of the openness of corruption to improve environmental pollution as $e^{18.42}$, $e^{16.79}$, and $e^{17.24}$, respectively. That is, when the level of the trade scale is higher than the critical value, corruption will reduce environmental pollution and improve environmental quality. From a regional perspective, using the per capita discharge of industrial wastewater as an example, the total trade scale level of the eastern region in 2017 was greater than the critical value. Further, the total trade scale level of the central and western regions was less than the critical value, indicating that, among the three major regions in China, only the eastern region's corruption has improved the environmental quality, while corruption in the central and western regions has exacerbated environmental pollution.

In Table 2, we can also determine the relationship between other variables and environmental pollution. The relationship between economic scale and environmental pollution presents an inverted U-shaped curve. This indicates that the EKC hypothesis is established in China and is in line with expectations and the increase in industrial structure indicators will aggravate environmental pollution. The possible reason for this scenario is that the speed and scale of the extensive economic development mode in the inspection interval have advantages over refined processing and the production of clean products, but the degree of pollution of the ecological environment is higher; technological progress has improved the environment quality. Environmental regulations are directly proportional to the severity of the pollution. It may be that high pollution fees indicate that the emission pollution is large scale. This might also reveal that companies prefer using high-emission and high-polluting production methods to avoid large costs of green transformation. The positive sign of FDI may be because the increase in pollution caused by lowering environmental standards to attract foreign investment, transfer of polluting industries, and expansion of domestic output caused by foreign investment exceeds the protective effect of foreign technology spillover effects on the environment, thereby exacerbating pollution.

5. Regional Heterogeneity Test

The following section verifies the conclusion that corruption will reduce environmental pollution when the level of trade scale is higher than the critical value and explains the phenomenon that the signs of technological progress in Table 2 are contrary to expectations.

Considering the obvious differences in location advantages and economic development levels, this section divides the sample data into three regions: east, west, and central, based on geographical location. Figure 1 shows the trend and comparison of the degree of corruption in the eastern, central, and western regions that have addressed using these methods.

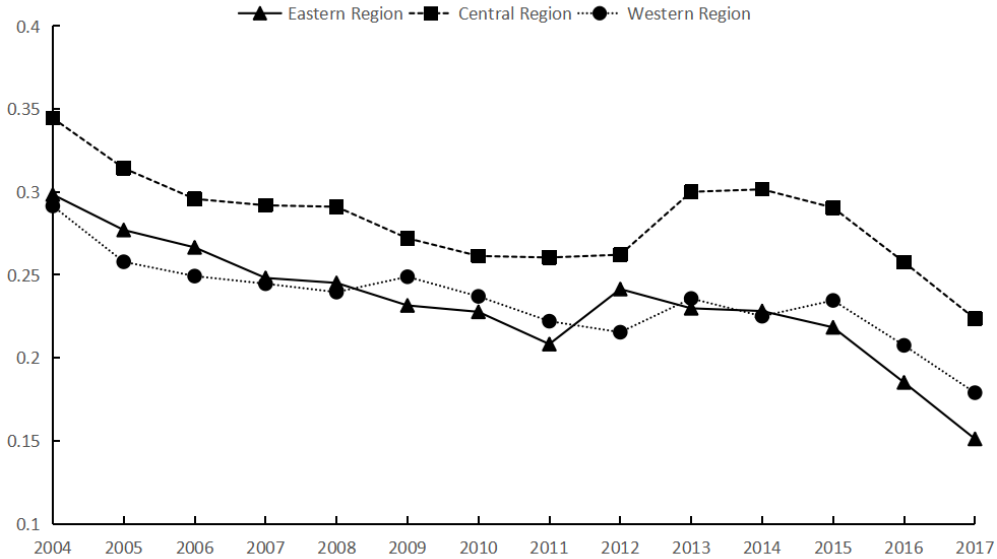

Figure 1. Time trends of corruption levels by region.

Figure 1 shows that, from 2003 to 2017, the degree of corruption in the eastern, central, and western regions in general tended to decline. In particular, there was a slight increase around 2013, but it declined at a faster rate in the following years. This shows that, after General Secretary Xi Jinping emphasized "putting power in the cage of the system" in 2013, the nation's strong-handed anti-corruption and power restriction investigations have been strengthened, and the warning effect has also achieved remarkable results. When comparing the eastern, central, and western regions, around 2013, the central group had the highest level of corruption, followed by the western and eastern groups; after 2013, the eastern group had the lowest corruption level. The reason may be that, in the decades before and after the 21st century, the reform of China's administrative system lags behind the pace of market economic reform. When the demand for production, circulation, and currency is expanding rapidly, if power approval and supervision procedures remain basically unchanged, corruption will become the center of economic circulation and resource distribution. However, the market economy pilot in the eastern region was earlier than that in the central and western regions, and it was ahead of the central and western regions in terms of openness, economic development level, use of advanced technologies, rational allocation of resources by the market, and efficiency of administrative reforms. Therefore, the degree of corruption is low, and "breaking away" from the credit system of economic circulation constructed by corruption in the past is easier and faster. However, the central and western regions are geographically close to the inland, based on traditional industries, and the market economy system is lagging behind in transition, and the development of factor and product markets is incomplete and unbalanced. Therefore, resource allocation in the two regions is not coordinated, the degree of corruption is high, and they rely heavily on corruption as a circulation channel.

Due to the differences in the degree of corruption in the east, middle, and west, the effects of corruption and trade-environmental pollution should be verified in different regions. Additionally, in Table 2, the empirical results of key variable lnt is not significant when per capita industrial wastewater discharge is used as an indicator, so there is a need to subdivide the regions for further exploration. Furthermore, the empirical results of regional heterogeneity for the other two indicators are unsatisfactory and not representative due to the varying implementation pressures of the policy environment in each region. Therefore,

the explained variable is selected as the per capita industrial wastewater discharge. Concurrently, T switches the total import and export volume per adult to test the robustness of the model. The model reference formula is the same as in (1), and the empirical test is presented in Table 3.

Table 3. Empirical results by region. (Explained variable: per capita industrial wastewater discharge.)

	Western Region	Central Region	Eastern Region
$L.\ln EP_{i,t-1}$	0.836 *** (19.32)	0.833 *** (11.48)	0.733 *** (10.76)
$\ln VG$	2.181 * (1.67)	2.791 * (1.79)	2.861 *** (3.27)
$\ln VG^2$	−0.137 ** (−2.07)	−0.137 * (−1.78)	−0.116 *** (−2.95)
$\ln IS$	0.304 (1.22)	0.146 (2.13)	0.276 ** (3.07)
$\ln TE$	0.338 *** (3.22)	−0.098 (−1.62)	−0.161 ** (−2.21)
$\ln EI$	0.105 ** (2.11)	−0.020 (−0.55)	−0.112 *** (−2.33)
$\ln T$	0.079 * (1.86)	0.026 (0.53)	−0.105 ** (−2.39)
$\ln CO$	0.212 *** (2.69)	0.135 *** (2.43)	−0.024 (−0.32)
$\ln F$	−0.003 (−0.15)	0.044 * (1.78)	0.042 *** (3.23)
cons	−9.797 (−1.53)	−14.308 * (−1.88)	−16.893 *** (−3.64)
AR(2)-*p* value	0.918	0.335	0.245
Sargan-*p* value	0.314	0.278	0.985

Note: *, **, and *** represent the significance level of 10%, 5%, and 1%, respectively. The values of the z-statistic are in parentheses. The AR(2) and Sargan test results are both *p* values. The null hypothesis is that no second-order serial correlation exists in the random error term of the first-order difference equation and the instrumental variables used are valid.

As shown in Table 3, in the eastern regions where the degree of openness is greater than the critical value, corruption improves the environmental quality. In the central and western regions where the degree of openness is less than the critical value, it deepens the degree of environmental pollution. This result confirms that a high degree of openness will indeed improve the pollution effect of corruption on the environment. First, when the expansion of trade scale magnifies the positive role of corruption as a deregulation mechanism in highly regulated economies with poor institutional environment, the advantages of corruption in improving the efficiency of resource allocation lead to outstanding contributions to environmental improvement. Second, the expansion of foreign trade may increase competition in the domestic market, reduce rent-seeking opportunities for enterprises, and restrain the negative effects of corruption that disrupt market order.

From the perspective of the impact of foreign trade on environmental pollution, it can be observed in Table 3 that foreign trade in the eastern region has improved the environmental pollution, which also verifies the results in Table 2. However, the foreign trade in the central and western regions aggravated the pollution. This may be because the geographical location and economic development level of the central and western regions are not as good as those of the eastern regions, and more pollution is produced during trade production and transportation. Moreover, factors such as the level of scientific research, human capital status, and investment status in these regions have not fully developed. This hinders or fails to fully use the technology spillover effect brought by trade and then fails to compensate for the scale effect and structural effect. The eastern region not only fully enjoys the benefits of technology spillover because of its earlier opening up but also the increase in people's income and consumption levels indicates that some critically

polluting products needed for life and production can be imported to meet demands, which also alleviates the scale effect of environmental pollution of export commodities. The effect of technological progress in the western region on environmental pollution in Table 3 is positive. However, the increase in the capital–labor ratio might promote environmental optimization, and the results in the eastern and central regions are in line with this expectation. This contradiction may be due to the differences in the level of human capital, R&D investment funds, and financing channels in various regions. In terms of exhaust gas emissions, the efficiency of the digestion and absorption of imported equipment, technology, and investment funds are different, and the ratio of capital to labor varies. To some extent, it also reflects the proportion of capital-intensive industries and does not directly reflect technological progress. The increase in the proportion of capital-intensive industries (e.g., metallurgical industry, petroleum industry, machinery manufacturing, and other heavy industries) is not conducive to environmental conditions. However, Table 3 shows that the expansion of FDI in various regions will pollute the environment. This indicates that, due to marketization, technological development speed, human capital quality, industrial structure optimization, insufficient supervision, and other factors, the utilization rate of environmental investment funds is limited. In Table 3, the signs of other explanatory variables are the same as the regression results in Table 2, proving the robustness of the model. It further proves that the current internal structure of China's industries has problems, such as low added value of products, high energy consumption, and high pollution, which have not been resolved. Hence, transforming the mode of economic development and realizing a new type of industrialization wherein the environment and economy develop in harmony are crucial.

6. Conclusions and Policy Recommendations

This paper examines the impact of corruption, foreign trade on environmental pollution in China, and how the impact of corruption on environmental pollution depends on the level of openness. After combing through the relevant literature, this paper initially conjectured that corruption leads to environmental degradation and that the level of trade openness determines the outcome of the pollution effect of corruption. In a subsequent empirical test, a dynamic panel data model is constructed, and a T $\times$ CO interaction term was introduced to extract the fraction of the environmental utility of corruption acting on foreign trade. A significant negative sign for T and a significant positive sign for CO suggest that foreign trade and corruption have opposite effects on the environment. Corruption exacerbates environmental pollution, while foreign trade plays a good role. The empirical test also indicates that the quadratic term of corruption is significantly positive, indicating that the positive role of corruption in improving the effect of trade on the environment may not be absolute but is temporary. While the regression coefficient of the interactive variables of corruption and foreign trade is significantly negative, this shows that higher levels of corruption play the bad role of trade in improving environmental pollution. That is, the increase in the level of corruption has weakened the positive effect of foreign trade on environmental improvement through channels such as the negative impact on per capita income, the creation of high trade barriers to hinder trade, the distortion of public investment, the reduction of environmental governance, and the efficiency of green innovation. On the other hand, the expansion of foreign trade can weaken the impact of corruption, which will provide good policy inspiration for improving the poor environmental impact of corruption. Moreover, at the regional level, this paper analyzes the current state of corruption in the east, middle, and west regions. It also finds that the relatively backward central and western regions have not enjoyed the benefits of opening up to the environment due to differences in the level of human capital, R&D investment funds, financing channels, and the efficiency of digestion and absorption of imported equipment, technology, and investment funds in each region. The results of this study allow us to provide policy recommendations on environmental governance from the perspective of trade openness and corruption.

First, China's current expansion of foreign trade will improve the degree of environmental pollution and even reduce the negative impact of corruption on the environment. Therefore, the overall expansion of opening up will improve the country's environmental pollution. Opening to the outside world under the new development pattern should focus on creating a high-level opening pattern to enhance the liberalization and facilitation of global resource utilization. Improving the investment and trade systems that are in line with advanced international rules and reducing the negative list of foreign business access would be desirable. While expanding opening up, strictly controlling the export structure and quality of foreign investment and formulating environmental entry thresholds in line with international environmental protection standards are necessary. Further, the scale of FDI should be expanded, and foreign capital inflows into high-tech and environmentally friendly industries must be encouraged and monitored.

Second, in the current imperfect system, moderate corruption can further play a beneficial role in the environment in trade by clearing the channels of resource flow and improving transaction efficiency, but is not a long-term solution. This aspect shows that, even after China has shifted from a planned economy to a market economy, drawbacks still restrict the vitality of market entities and the flow of factors. That is, in the process of the government's macro-control of the economy, the excessively high position of administration in the market and the insufficient development of the factor market make some private enterprises exert their competitive advantages through collusion between government and business. Hence, China is required to further deepen market-oriented reforms; improve the fair competition review system; promote the reform of the household registration system, market-oriented allocation of capital market factors, and market-determining factor price mechanisms; and establish open, fair, and transparent market rules. The government should strengthen exchanges with enterprises, especially those in the non-public sector of the economy, reinforce the ability to identify political achievements, and create a new "close" and "clean" political and business environment.

Third, in recent years, China's investment in environmental governance has not been efficient in improving pollution, and the collection of pollution fees has not played a significant role because of issues such as renovation costs. Therefore, to fully understand the government's active role in environmental governance, the concentration and precision of pollution treatment should be increased. Proactive measures would be subdividing the status quo of pollution control in different industries, companies, and regions, adjusting the scale and structure of governance input, carefully choosing environmental regulation methods, optimizing the spatial layout of wastewater treatment facilities, and innovating third-party public service supply methods, such as service outsourcing. Simultaneously, the efficiency loss caused by diminishing returns to scale by improving the system, promoting technological progress, and introducing new pollution control technologies and management models must be reduced.

Fourth, in general, China's environmental pollution has intensified as the proportion of the secondary industry's GDP increases. This development shows problems such as insufficient advancement of the industrial structure, high production consumption, and high pollution within the overall industry and the secondary industry. Changing traditional production methods, enhancing product added value and technological content, and developing high-end industrial chains are the focus of realizing new industrialization. In this process, the government must grasp the new characteristics of the modern industrial system, use the means of enhancing innovation-driven development capabilities, continuously optimize the industrial development institutional environment, and attract high-quality talents, high-innovation platforms, high-quality parks, and high-flow funds to maximize the technology spillover effects of opening up. Subsequently, the government should support enterprises to enhance its independent development capabilities and form the prosperity and development pattern of high-tech industry and strategic emerging industries vigorously. This has long-term benefits for coordinated environmental and economic development.

7. Limitations and Improvements

The limitations of this paper are mainly in the following areas. Firstly, due to the hidden nature of regional corruption and the limitations of data availability, there are limited indicators to measure corruption that could be further refined. Future research could use and develop different indicators to provide a more comprehensive picture of the level of regional corruption. More data could be collected on cases of government corruption, efficiency in the use of government funds, and inactivity of civil servants. With the development of information technology, the use of big data methods to collect corruption data can be further explored. Meanwhile, this paper analyzes the impact of corruption on environmental pollution mainly from the perspective of governance. Moreover, regional corruption is only one of the factors that affects regional environmental governance. Carbon governance not only depends on the quality of government systems, but is also influenced by the level of economic development and the level of technology. In addition, uncertainty in the external environment and major emergencies such as epidemics can also affect the strength and effectiveness of environmental governance. Scholars Su [53] and Kartal [54] have also clarified the important role of better political environment in their studies. Therefore, in future research, a more comprehensive analysis can be conducted by considering the impact of institutional quality, government policies, economic factors, and environmental factors on carbon governance.

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